

Applied Research Methods: Assessment Two

Major Project Research Proposal

1	Name:	Hardiksinh Ajaysinh Parmar
2	Course:	MSc Geographic Information System
3	Project title:	EVALUATING THE EFFICIENCY OF EXISTING WASTE MANAGEMENT STRATEGIES BY USING GIS IN SHEFFIELD
4	Aims and objectives/research questions	
To evaluate Sheffield's present waste management methods for efficacy. Data on garbage creation, collection, disposal, and recycling in the city would be collected for this assessment. To ascertain the existing state of waste management efficiency and to pinpoint any potential improvement areas, the data would be analysed. Numerous measures, including recycling rates, landfill usage, and waste diversion rates, might be used in this analysis. Map and analyse waste management data using GIS technology. The spatial distribution of garbage generation, collection, and disposal can be visualised using GIS technology, and any spatial patterns or trends can be found. This knowledge can then be applied to create waste management plans and regulations that are more focused on different areas of the city. Create suggestions for bettering Sheffield's waste management. These suggestions would consider elements including price, sustainability, and community engagement. The introduction of more recycling facilities in residential areas, the implementation of composting initiatives, and the creation of educational initiatives to raise awareness of the value of trash reduction and recycling are a few examples of recommendations. Help other cities and communities throughout the world, as well as Sheffield, establish waste management policies that are more effective and sustainable. The study's conclusions and suggestions could serve as a guide for other towns and cities looking to enhance their trash disposal procedures. To raise public knowledge of the value of effective waste management and to promote community involvement in the creation of more environmentally friendly waste management methods. This could be accomplished by creating outreach and education initiatives that raise public awareness of recycling and trash reduction. The creation of community recycling facilities and other community-based initiatives could help promote community participation in waste management techniques.		

5	Research rationale <i>The provision of a clear rationale ('literature review') for the study which highlights its timeliness and discusses the contribution your research will make to existing knowledge of theory and practice on your topic. This should cover the main themes of your research, briefly summarise existing knowledge, including any competing perspectives, and focus on what your research will add to this. This should draw predominantly on academic texts, but some additional good quality practice material is also acceptable</i>
	1. Geographic Information Systems (GIS) are used in waste management, and the research project "The Assessment of GIS Use in Waste Management" examines this use. The study's authors sought to assess the utility of GIS in waste management and to determine both the advantages and drawbacks of its application. • The study used a questionnaire survey to collect information from waste management industry specialists with GIS experience. The questions in the survey centred on the many GIS applications used in waste management, the advantages and disadvantages of using GIS, and the general efficacy of GIS in controlling waste. • According to the study's conclusions, GIS is a useful tool for managing waste, especially when it comes to planning and decision-making. The study also discovered that using GIS for waste management provides a number of advantages, including better data management, higher effectiveness and cost-effectiveness, and better stakeholder engagement and communication. • However, the study also noted some drawbacks to the application of GIS in waste management, such as the expense of doing so, the requirement for qualified personnel to operate and maintain GIS systems, and problems with data quality. (Cristina Iacoboaea & Mihaela Aldea, 2011) 2. The article "Using GIS in Solid Waste Management Planning: A case study for Aurangabad, India" proposes a GIS-based model for planning waste management strategies in the city of Aurangabad, India. The model takes into consideration various aspects, such as the convenience and inconvenience of existing bin locations, segregation of waste into two categories, revenue generation, and public participation. • The study begins by analyzing the location of existing waste bins in the area and proposes the provision of recyclable waste collection bins in areas that generate recyclable waste. The proposed model is designed to be cost-effective and takes into consideration budget constraints. It also emphasizes revenue generation from both categories of waste to support the waste management system. • The study highlights the importance of public participation in waste management strategies and suggests ways to effectively involve the public in the process. The model proposed in the study is flexible and can be upgraded as per situational changes or preferences. The study also emphasizes the need to consider various factors, such as data availability and situational constraints, when planning waste management strategies. In general the study provides valuable insights into using GIS for solid waste management planning and highlights the importance of considering various factors in planning waste management strategies. (Ahmed, 2006) 3. The article you provided discusses a study that focused on using Geographic Information Systems (GIS) as a tool for selecting environmentally friendly and

healthy landfill sites in Harar city. The study concluded that GIS is an effective tool for selecting such sites, and it introduced the estimation of waste volume and landfill size calculation as an independent factor for site selection, which is unique compared to other studies.

- One of the key strengths of the study was the use of the opinions of experienced experts to rank each criterion, which is an important but often neglected part of landfill site selection studies. By incorporating expert opinions, the study was able to more accurately and comprehensively evaluate the different criteria involved in selecting a landfill site.
 - The study's results have important implications for decision-making in Harar city. Policymakers can use the future trend of waste generation calculated in the study as a baseline for developing waste management policies and strategies. The thematic maps generated during the study can also serve as a valuable resource for identifying and studying related problems, while the methodology used in the study can be adopted in other areas facing similar sanitary landfill siting problems. (Asefa et al., 2021)
4. The ArcGIS Network Analyst tool was used in this study to create an optimisation for the collection and transportation of garbage in three Jordanian cities. In order to determine the best routes, the study created and analysed three scenarios with various adjustments to the working hours, collection method, transportation gear, and transfer locations. The findings revealed that combining the S1 and S2 scenarios resulted in significant cost and labour savings, as well as a reduction in vehicle running time and other advantages for CO2 emissions, fuel efficiency, and vehicle maintenance.
- The study came to the conclusion that decision-makers can effectively solve the complicated challenge of urban trash management by using simple heuristic methods and GIS technology. The GIS-based enhanced scenarios offer a numerical and objective approach to answer the operational question of how to collect and dispose of solid waste, rather than relying on biased perceptions of drivers. (Hemidat et al., 2017)
5. The article titled "Optimal municipal solid waste collection system design using GIS: a case study" presents a study on the design of an optimal municipal solid waste collection system using Geographic Information System (GIS) technology. The study was conducted in the city of Istanbul, Turkey, and aimed to determine the optimal collection routes and frequencies for the waste collection trucks.
- The authors used GIS technology to develop a database of the city's streets, buildings, and waste collection points. They then used the database to create optimal collection routes based on factors such as the location of waste collection points, the amount of waste generated, and the distance between collection points.
 - The study compared the performance of three different waste collection scenarios: the current waste collection system, a hypothetical centralized waste collection system, and an optimal waste collection system based on GIS analysis. The results showed that the optimal waste collection system based on GIS analysis was more efficient and cost-effective than both the current system and the centralized system.
 - The study concludes that GIS technology can be a useful tool in designing optimal waste collection systems. The use of GIS allows for a more accurate and

comprehensive analysis of the waste collection process, which can lead to significant cost savings and environmental benefits. (Leao et al., 2001)

6. The article discusses how the environmental impact of demolition waste management (DWM) is often overlooked, despite its importance in sustainability and urban planning. To address this issue, the authors propose a new evaluation model that combines Geographic Information Systems (GIS) and system hybrid simulation to assess the environmental impact of DWM at the city scale.
 - The article presents an empirical analysis of the model in Shenzhen, China, where the authors demonstrate the efficacy of the model. They conclude that their model is a promising tool to assess the environmental impact of DWM in urban areas and can help decision-makers evaluate and improve DWM policies and practices.
 - In summary, the article highlights the importance of considering the environmental impact of DWM and proposes a new evaluation model that integrates GIS and system hybrid simulation to assess the impact at the city scale. The model is demonstrated to be effective in an empirical analysis in Shenzhen, China, and can be a useful tool for decision-makers in evaluating and improving DWM policies and practices. (Ding et al., 2022)
7. The article is about a study that evaluated operational municipal solid waste incinerators in the UK, which are facilities that burn waste to produce energy. The study looked at four specific waste incinerator plants and analyzed their supply chain, including factors such as technical, financial, and operational data. The authors suggest that waste incinerators can benefit from adopting best business practices, and that choosing the right strategies can be a significant challenge for these plants. The study is important because it provides insight into the operations and challenges faced by waste incinerators in the UK, which can help inform future decision-making related to waste management. By understanding the issues and difficulties faced by waste incinerators, policymakers and waste management professionals can make more informed decisions about how to manage waste in an efficient, effective, and environmentally responsible manner. (Nixon et al., 2013)
8. The use of geographic information systems for solid waste management is the topic of this chapter. GIS is an effective technology that, by providing more and better information, can help to reduce value conflicts among parties with competing interests. In essence, establishing a sustainable waste management method is difficult, time-consuming, and complex, and decision-makers are usually faced with competing demands.
 - Waste generation is on the rise globally, and the situation is worse in underdeveloped nations due to a lack of infrastructure, inadequate funding, and political factors. Integrated solid waste management is a popular sustainable waste management technique used to address the excessive waste creation and various SWM issues. Implementing ISWM is a challenging and laborious task. . Following the difficulties in ISWM, numerous researchers developed numerous innovations, including, but not limited to, mathematical modelling and computer-based modelling. GIS is a computer-based spatial analysis technique used in SWM that enables decision-makers to combine their preferences and the alternatives to make better decisions. This chapter covered the fundamentals and concepts of GIS, what multicriteria decision making is, how to use it with GIS, and how to use the GIS-

based MCDA approach in SWM. Additionally, details on how to use GIS for planning waste disposal (landfill) and waste collection optimisation (routing and waste bin allocation) were provided. Additionally, references for further reading were provided to support the fundamental ideas of the development of the MCDA method based on GIS. (Asefa et al., 2022)

9. Solid waste management is a critical issue faced by municipalities and local governments worldwide. The increasing population and industrial development have led to a significant increase in waste generation, creating challenges for waste management systems. Kharagpur is one such city in India that faces a severe waste management problem due to its population growth and industrial development.
 - To address the issue of waste management in Kharagpur, there is a need to identify cost-effective and ecologically sound alternative landfill sites. The selection of landfill sites involves the consideration of various criteria such as environmental impact, accessibility, proximity to the source of waste, and potential for future expansion.
 - To select the most appropriate landfill sites, different multi-criteria weights methods can be adopted. These methods involve assigning weights to each criterion based on their relative importance and then evaluating the potential sites based on these weighted criteria. The results of this analysis would be helpful in developing an efficient solid waste management system for Kharagpur. A well-planned waste management system can not only reduce the environmental impact of waste but also contribute to the sustainable development of the city. Therefore, it is crucial for the local authorities to take effective measures to manage solid waste efficiently and sustainably. (Dolui & Sarkar, 2021)
10. The integration between GIS software and multicriteria evaluation (MCE) can be a powerful tool in evaluating and recommending future landfill sites. This approach has been used in the case of the West Bank to identify the most suitable alternative locations for landfill siting in limited and disputed lands.
 - The study has provided evidence of how unsuitable the existing landfills and dumping sites are in the West Bank. MCE has been used to provide a hierarchy of criteria and sub-criteria with weights and ranks to enhance the GIS modeling of these criteria. The criteria used in this study include land vulnerability to damage and water contamination, streams, landscapes, natural reserves, biodiversity, wells, Jordan River, agricultural value of land, slope, roads, Palestinian residential built-up areas, Dead Sea, political disputes, Israeli settlements, political classification of land (OSLO accord A, B, and C), springs, land ownership, closed areas by the Israeli wall, archaeological areas, and existing dumping sites.
 - The weighted criteria, along with sub-criterion ranks, have been overlaid using the "weighted overlay" tool in ArcMap (GIS) to obtain the results. The suitable locations for the West Bank landfills represent about 6% of the West Bank, while all existing regional (and planned) landfills are located in the restricted areas based on this study.
 - Therefore, it is strongly recommended to reallocate existing and future planned landfills to more preferable sites. The study suggests bringing this issue onto future negotiation agendas to avoid the adverse impact on the environment and public health. The approach used in this study can be adopted in other regions to evaluate and recommend suitable landfill sites to ensure efficient and sustainable waste management. (Helal, 2022)

11. The research paper titled "Urban Solid Waste Management Using GIS Technique: A Case Study on Mohammadpur Thana at Dhaka of Bangladesh". The paper discusses the use of GIS (Geographic Information System) technique in managing urban solid waste in Mohammadpur Thana, a district in Dhaka, Bangladesh.
 - The study focuses on identifying suitable landfill sites for the district using GIS. The research methodology involved collecting primary data through surveys and field visits and secondary data from various sources such as maps, satellite images, and reports. The collected data were analyzed using the GIS technique to develop a spatial database that includes information on the waste generation rate, waste collection, and transportation routes, and landfill sites.
 - The study found that the existing landfill site in Mohammadpur Thana was not suitable due to its proximity to residential areas, water bodies, and potential environmental hazards. The research suggested identifying a new landfill site in a more suitable location to manage the waste efficiently and sustainably.
 - The paper highlights the importance of using GIS technique in urban solid waste management to analyze and visualize data spatially, allowing for informed decision-making regarding waste management. The approach used in this study can be useful in other regions to develop effective and sustainable waste management systems. (Syed et al., 2009)

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Methodology and methods

An explanation of how your topic area has been researched in the past and how this has informed your overall research strategy (methodology) and research methods. This explanation should conclude with a summary of your chosen methodology and methods and demonstrate how your methods will provide the data you need to answer your research aims / questions. This needs to include a consideration of your data analysis.

Data collection on current practises is the first step in enhancing waste management procedures. Gathering data on garbage generation, collection, disposal, and recycling may be a part of this data collection. Surveys, site visits, and other methods may be used to gather the data. Then, using this data, a thorough understanding of the waste management procedures in use may be created.

The data is then mapped using GIS technology in the following stage. The spatial distribution of waste generation, collection, and disposal can be seen using this mapping. It can also be utilised to spot any spatial trends or patterns that could be present. The mapping can be used to pinpoint places with high rates of trash production, insufficient garbage disposal infrastructure, and high recycling potential.

The third stage is to examine the information to find areas that could use improvement. The waste management system's inefficiencies can be found, and places where new trash reduction or recycling programmes can be created, can also be found through this research. GIS technology offers capabilities for data analysis and visualisation, which might help with this study.

The creation of sustainable waste management strategies is the fourth step. Based on the data gathered and the analysis done, these strategies can be established. The implementation of recycling programmes, the creation of additional trash disposal facilities, and the introduction of new waste reduction programmes are some possible strategies.

Cost, sustainability, and community involvement should all be taken into account while developing these methods.

The monitoring and evaluation of the efficacy of the waste management methods' execution constitutes the fifth phase. This entails monitoring the results of the methods put into place and assessing how well they work to decrease waste production, increase recycling, and enhance trash disposal procedures. GIS technology can be used to evaluate how waste management practises have changed over time.

In summary, utilising GIS technology is crucial for enhancing waste management procedures. It enables the gathering and mapping of data, the analysis of the data to pinpoint areas that need improvement, and the creation of long-term waste management plans. To guarantee the success of these tactics, it is also essential to track and assess how well they are being implemented. We can lessen the detrimental effects on the environment and encourage a more sustainable future by improving waste management procedures.

7	Research ethics <i>A discussion of any potential ethical issues you anticipate in your research, how you intend to address them and how this has informed your chosen methodological approach and research methods.</i>
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The use of secondary data and satellite imagery in the research allows for a non-invasive approach, mitigating any potential risks or harm to human participants, and demonstrates a commitment to upholding ethical principles in research.

8	Have you completed the appropriate UREC form? You must submit this as an appendix alongside your proposal.	Yes
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9	References
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Investigation of groundwater quality of Ramjannagar Union of Shyamnagar Upazila at Satkhira District, Bangladesh View project Climate Finance Transparency Mechanisms View project